Flipping Coins Again. Suppose we have a coin which flips heads with probability p. Recall that the expected number of flips until one heads is $1/p$. The derivation of this fact is as follows: let X be the random variable counting the number of flips of the above coin until we get 1 heads. Then,

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{i=1}^{\infty} i \cdot \Pr[X = i] && \text{(definition of Expectation)} \\
&= \sum_{i=1}^{\infty} i \cdot \Pr[\text{first } (i-1) \text{ flips are tails}] \cdot \Pr[i\text{th flip is heads}] \\
&= \underbrace{\sum_{i=1}^{\infty} i \cdot (1-p)^{i-1}}_{=\frac{1}{(1-(1-p))^2}} \cdot p = 1/p,
\end{aligned}$$

where we used the fact that for every $x \in [0, 1)$, $\sum_{i=1}^{\infty} i x^{i-1} = \frac{1}{(1-x)^2}$.

1. Derive the expected number of flips until we get 2 heads using a similar calculation to the one above. Hint: $\sum_{i=2}^{\infty} i \cdot (i-1) x^{i-2} = \frac{2}{x^3}$.
2. Is there an easier way to find the expected number of flips until we get 2 heads using linearity of expectation? Hint: Define extra random variables X_1 and X_2 such that $X = X_1 + X_2$. Hinthint: The coin doesn't remember its past flips.

Flipping Coins Again Again. Suppose we have a fair coin which flips heads with probability 1/2. In expectation, how many flips are required to get 2 heads in a row? How about 3 heads in a row? Hint: the coin does not remember its past flips.

John Pork is Calling. You answer, and suddenly you're on the gameshow, "Who Wants to Be the Sigma?" Three doors stand before you. Behind one is infinite rizz. Behind the other two is  who whispers "paws at you" and permanently deletes your aura. You pick Door 1.

The host, the Rizzler himself, knows what's behind every door. He always opens a door you didn't pick, always reveals a "paws at you" kitten, and always lets you switch.

He opens Door 3.  *paws at you*

John Pork calls again. "Last chance, chat. Stay with Door 1, or switch to Door 2?" Which move maximizes your rizz: locking in or switching, and what's your chance of winning?